

CLASS TEST

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT) — THIRD SEMESTER, DECEMBER 2023

Paper Code: ES-201

Subject: Computational Methods

Time: 1 Hr.

Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Marks	CO
1(a)	Find u_1, u_2, u_3 and u_4 initially for $u_{xx} + u_{yy} = 0$ with the help of the figure:	[3]	CO4
1(b)	Solve the system of equations by Gauss elimination method: $2x - y + 3z = 9, \quad x + y + z = 6, \quad x - y + z = 2$	[4]	CO3
1(c)	Given $dy/dx = (y - x) / (y + x)$, with the initial condition $y = 1$ at $x = 0$. Find y for $x = 0.1$ and $x = 0.2$ using Euler's method.	[3]	CO4
2	Solve the following system of equations by Cholesky's method: $x + y + 3z = 6, \quad x + 5y + 5z = 20, \quad 3x + 5y + 19z = 106$	[10]	CO3
3	By using Runge-Kutta fourth order method to find y for $x = 0.1$ in step of $h = 0.1$ if $dy/dx = x + y^2$, Given that $y(0) = 1$.	[10]	CO4
4	Find the value of $u(x, t)$ satisfying the parabolic equation $\partial u / \partial t = 4 (\partial^2 u / \partial x^2)$ and the boundary conditions $u(0, t) = 0, u(8, t) = 0$, and $u(x, 0) = 4x - \frac{1}{2} x^2$ at the points $x = i$ ($i = 0, 1, 2, 3, 4, \dots$) and $t = (1/8) j$ ($j = 0, 1, 2, 3, 4, 5$).	[10]	CO4

**** End of Question Paper ****